King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Final Exam Term 231

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. If $f(x) = \begin{cases} x^2 + 1 & x \leq 2 \\ 4x - 3 & x > 2, \end{cases}$ then $f'(2) =$

(a) 4

(b) 1

(c) 2

(d) 0

(e) does not exist

2. If y is defined implicitly as a differentiable function of x by the equation $x^3y^3 - y = x$, then $\dfrac{dy}{dx} =$

(a) $\dfrac{1 - 3x^2y^3}{3x^3y^2}$

(b) $\dfrac{x - y}{x^2y^2 - 1}$

(c) $\dfrac{3x^2y^3}{3x^3y^2 - 1}$

(d) $\dfrac{1 - 3x^2y^3}{3x^3y^2 - 1}$

(e) $\dfrac{1 - x^2y^3}{x^3y^2 - 1}$

3. The slope of the tangent line to the graph of $f(x) = \dfrac{1}{x}$, $x > 0$, that passes through the point $(1, -3)$ is equal to

(a) $-\dfrac{9}{4}$

(b) $-\dfrac{25}{4}$

(c) -4

(d) -25

(e) -9

4. The differential dy of the function $y = x(1 - \cos x)$ is

(a) $dy = (1 + \cos x - x \sin x)\, dx$

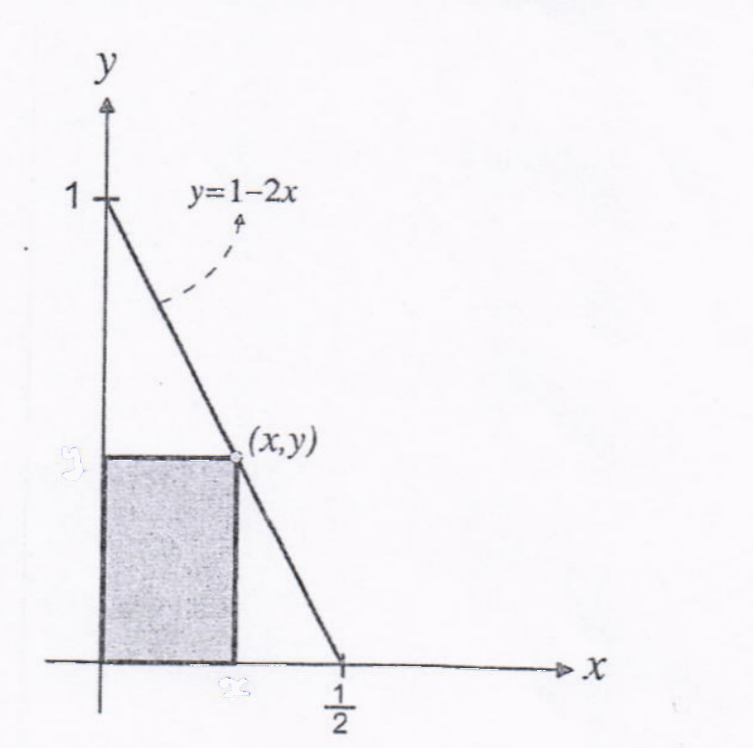
(b) $dy = (1 - x \sin x)\, dx$

(c) $dy = (1 - \cos x + x \sin x)\, dx$

(d) $dy = (\cos x - x \sin x)\, dx$

(e) $dy = (x - \cos x + x \sin x)\, dx$

5. A rectangle is to be constructed in such away that it will be bounded by the x-axis, y-axis, and the graph of the line $y = 1 - 2x$ (as shown in the figure). What is the maximum possible area of such a rectangle?



(a) $\frac{1}{4}$

(b) $\frac{1}{8}$

(c) $\frac{5}{3}$

(d) $\frac{2}{3}$

(e) $\frac{3}{4}$

6. If $f(x) = \begin{cases} (e^x + 2x)^{1/x} & x \neq 0 \\ c & x = 0 \end{cases}$ is continuous at $x = 0$, then $c =$

(a) e

(b) e^3

(c) e^2

(d) 1

(e) e^{-2}

7. $\lim_{x \to (1/2)^+} \dfrac{x}{2x - 1} =$

(a) ∞

(b) $\dfrac{1}{2}$

(c) $-\infty$

(d) 0

(e) -1

8. The slant asymptote of the graph of the function $f(x) = \dfrac{4x^3 + x}{2x^2 + 1}$ is given by

(a) $y = \dfrac{1}{2}x + 1$

(b) $y = 2x$

(c) $y = 2x - \dfrac{1}{2}$

(d) $y = 4x$

(e) $y = -2x + 1$

9. If the function $f(x) = 2x + \dfrac{1}{x}$ has a relative maximum at the point (a, b), then $a =$

(a) $\dfrac{1}{2}$

(b) $-\dfrac{1}{2}$

(c) $-\dfrac{\sqrt{2}}{2}$

(d) 0

(e) $\dfrac{\sqrt{2}}{2}$

10. The slope of the tangent line to the graph of $f(x) = \dfrac{5}{x^3 - 2}$ at the point $\left(-2, -\dfrac{1}{2}\right)$ is equal to

(a) $-\dfrac{3}{10}$

(b) $-\dfrac{5}{2}$

(c) $-\dfrac{15}{2}$

(d) -1

(e) $-\dfrac{3}{5}$

11. Which one of the following statements is FALSE about the graph of the function $g(x) = 2x \ln x$?

(a) The line $x = 0$ is a vertical asymptote for the graph.

(b) The graph is concave up on $(0, \infty)$.

(c) The graph has one local minimum.

(d) The graph is increasing on (e^{-1}, ∞).

(e) The graph has no horizontal asymptote.

12. If $\sinh x = -\dfrac{1}{2}$, then $\tanh x =$

(a) $-\dfrac{\sqrt{3}}{3}$

(b) $-\sqrt{5}$

(c) $\dfrac{\sqrt{5}}{5}$

(d) $\dfrac{\sqrt{5}}{2}$

(e) $-\dfrac{\sqrt{5}}{5}$

13. If $f(x) = \dfrac{x^2\sqrt{3x-2}}{(x+1)^2}$, then $f'(1) =$

(a) $\dfrac{5}{6}$

(b) $\dfrac{3}{4}$

(c) $\dfrac{5}{4}$

(d) $\dfrac{5}{8}$

(e) $\dfrac{3}{8}$

14. $\int (2x^3 - \sec^2 x)\, dx =$

(a) $2x^4 - \tan x + c$

(b) $\frac{1}{2}x^4 - \tan x + c$

(c) $\frac{1}{2}x^4 + \tan x + c$

(d) $2x^4 - \cot x + c$

(e) $x^4 + \tan x + c$

15. $\lim_{x \to 3^+} \left(\frac{18}{x^2 - 9} - \frac{x}{x - 3} \right) =$

(a) $-\frac{1}{6}$

(b) ∞

(c) $-\frac{3}{2}$

(d) 1

(e) 0

16. The tangent line approximation of the function $f(x) = \sqrt{x} + \frac{2}{\sqrt{x}}$ at the point $(1, 3)$ is

(a) $y = -\frac{1}{2}x + \frac{7}{2}$

(b) $y = -\frac{3}{2}x + \frac{9}{2}$

(c) $y = -x + 4$

(d) $y = \frac{1}{2}x + \frac{5}{2}$

(e) $y = -\frac{1}{3}x + \frac{10}{3}$

17. The measurement of the edge of a cube is found to be $8\,cm$, with a possible error of $0.02\,cm$. Using differentials, the possible propagated error in computing the volume of the cube is approximately equal to

(a) $1.92\,cm^3$

(b) $2.54\,cm^3$

(c) $3.33\,cm^3$

(d) $1.28\,cm^3$

(e) $3.84\,cm^3$

18. The graph of $f(x) = x\sqrt{x+2}$ is concave upward on the interval(s)

(a) $(-2, \infty)$

(b) $\left(-\frac{8}{3}, -2\right)$

(c) $\left(-\infty, -\frac{8}{3}\right)$ and $(-2, \infty)$

(d) $(-\infty, \infty)$

(e) $\left(-\frac{8}{3}, \infty\right)$

19. If $f(x) = 3x^4 - 4x^3$ has an absolute minimum at the point (a, b) on the interval $[-1, 2]$, then $ab =$

(a) -1

(b) 10

(c) 0

(d) 32

(e) -7

20. If (a, b) is the point on the curve $y = \sqrt{x}$ that is closest to the point $(1, 0)$, then $ab =$

(a) $2\sqrt{2}$

(b) $\frac{\sqrt{2}}{4}$

(c) $\frac{\sqrt{2}}{2}$

(d) $\sqrt{2}$

(e) $\frac{\sqrt{2}}{3}$

21. If (a, b) is the interval on which $f(x) = \frac{\ln(2-x)}{\sqrt{2x-3}}$ is continuous, then $ab =$

(a) 2

(b) 4

(c) $\frac{5}{2}$

(d) $\frac{9}{2}$

(e) 3

22. If $f(x) = \frac{1}{2}\ln(\sinh(x^2))$, then $f'(x) =$

(a) $x\coth(x^2)$

(b) $\frac{1}{2}x\,\mathrm{csch}(x^2)$

(c) $\frac{1}{2}x\cosh(x^2)$

(d) $2x\,\mathrm{sech}(x^2)$

(e) $x\tanh(x^2)$

23. $\lim_{x\to\infty}\frac{(\ln x)^2}{x^3} =$

(a) $\frac{2}{9}$

(b) 0

(c) $\frac{2}{3}$

(d) 1

(e) ∞

24. An equation of the tangent line to the graph of $f(x) = -2x^4 + 5x^2 - 3$ at the point $(1, 0)$ is given by

(a) $y = 2x - 3$

(b) $y = 3x - 3$

(c) $y = -2x + 2$

(d) $y = 2x - 2$

(e) $y = \frac{1}{2}x - \frac{1}{2}$

25. Which one of the following statements is TRUE?

(a) If $h(x) = f(x)g(x)$ and $f'(c) = g'(c) = 1$, then $h'(c) = 1$.

(b) If $f'(x) = g'(x)$, then $f(x) = g(x)$.

(c) If $y = (x+1)(x+2)(x+3)$, then $\frac{d^3y}{dx^3} = 0$.

(d) If $f(x) = \pi^3$, then $f'(x) = 3\pi^2$.

(e) If $f(x) = \frac{1}{g(x)}$ and $g(c) = g'(c) = -1$, then $f'(c) = 1$.

26. The function $f(x) = x^3 - 6x^2 + 10$ is decreasing on the interval(s)

(a) $(0, 4)$

(b) $(-\infty, 4)$

(c) $(-\infty, 0)$ and $(4, \infty)$

(d) $(0, \infty)$

(e) $(-\infty, \infty)$

27. If $f''(x) = x^{-3/2}$, $f'(4) = 2$, $f(1) = 1$, then $f(9) =$

(a) 9

(b) 17

(c) 11

(d) 15

(e) 13

28. The number of inflection points of the graph of $f(x) = e^{-x^2}$ is

(a) 4

(b) 1

(c) 0

(d) 3

(e) 2

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	Left and right difference quotients both give 4, so $f'(2) = 4$.
2	**(d)**	$3x^2y^3 + 3x^3y^2y' - y' = 1$, so $y' = \frac{1-3x^2y^3}{3x^3y^2-1}$.
3	**(e)**	Tangency at $t = \frac{1}{3}$ ($3t^2 + 2t - 1 = 0$), slope $-1/t^2 = -9$.
4	**(c)**	$y' = (1 - \cos x) + x \sin x$, so $dy = (1 - \cos x + x \sin x)\, dx$.
5	**(b)**	$A = x - 2x^2$, $A' = 0$ at $x = \frac{1}{4}$, $A_{\max} = \frac{1}{8}$.
6	**(b)**	$\ln y \to \lim \frac{e^x+2}{e^x+2x} = 3$, so $c = e^3$.
7	**(a)**	Numerator $\to \frac{1}{2} > 0$, denominator $\to 0^+$, so the limit is ∞.
8	**(b)**	$f = 2x - \frac{x}{2x^2+1}$, so the slant asymptote is $y = 2x$.
9	**(c)**	$f' = \frac{2x^2-1}{x^2}$ changes $+ \to -$ at $x = -\frac{1}{\sqrt{2}} = -\frac{\sqrt{2}}{2}$.
10	**(e)**	$f'(x) = \frac{-15x^2}{(x^3-2)^2}$, $f'(-2) = -\frac{60}{100} = -\frac{3}{5}$.
11	**(a)**	$\lim_{x\to 0^+} 2x \ln x = 0$, so $x = 0$ is not a vertical asymptote (false statement).
12	**(e)**	$\cosh x = \frac{\sqrt{5}}{2}$, so $\tanh x = -\frac{1}{\sqrt{5}} = -\frac{\sqrt{5}}{5}$.
13	**(d)**	Log differentiation: $f'/f = 2 + \frac{3}{2} - 1 = \frac{5}{2}$ at 1, $f(1) = \frac{1}{4}$, so $f'(1) = \frac{5}{8}$.
14	**(b)**	$\int 2x^3 = \frac{1}{2}x^4$, $\int \sec^2 x = \tan x$: $\frac{1}{2}x^4 - \tan x + c$.

Q	Ans	Brief explanation
15	**(c)**	Combine: $\frac{-(x+6)(x-3)}{(x-3)(x+3)} \to -\frac{9}{6} = -\frac{3}{2}$.
16	**(a)**	$f'(1) = -\frac{1}{2}$, $y = 3 - \frac{1}{2}(x-1) = -\frac{1}{2}x + \frac{7}{2}$.
17	**(e)**	$dV = 3x^2 dx = 3(64)(0.02) = 3.84\,cm^3$.
18	**(a)**	$f'' = \frac{3x+8}{4(x+2)^{3/2}} > 0$ for all $x > -2$.
19	**(a)**	Values $0, -1, 7, 16$: minimum -1 at $x = 1$, so $ab = -1$.
20	**(b)**	Minimise $(x-1)^2 + x$: $x = \frac{1}{2}$, $y = \frac{1}{\sqrt{2}}$, $ab = \frac{\sqrt{2}}{4}$.
21	**(e)**	Domain $\left(\frac{3}{2}, 2\right)$, so $ab = 3$.
22	**(a)**	Chain rule: $\frac{1}{2} \cdot \frac{\cosh(x^2)}{\sinh(x^2)} \cdot 2x = x\coth(x^2)$.
23	**(b)**	Two applications of L'Hopital give $\frac{2}{9}\lim \frac{1}{x^3} = 0$.
24	**(d)**	$f'(1) = 2$, so $y = 2(x-1) = 2x - 2$.
25	**(e)**	Only the reciprocal-rule statement holds: $f'(c) = -\frac{g'}{g^2} = 1$.
26	**(a)**	$f' = 3x(x-4) < 0$ exactly on $(0, 4)$.
27	**(b)**	$f' = -2x^{-1/2} + 3$, $f = -4\sqrt{x} + 3x + 2$, so $f(9) = 17$.
28	**(e)**	$f'' = -2e^{-x^2}(1 - 2x^2)$ changes sign at $x = \pm\frac{\sqrt{2}}{2}$: two points.